



# ELECTRÓNICA DIGITAL III

## CLASE 3

### Módulo Interno NVIC

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# Módulo SCS

Dentro del mapa de memoria se encuentra una región denominada *SCS (System Control Space)* encargada del control de diversas funcionalidades del sistema:

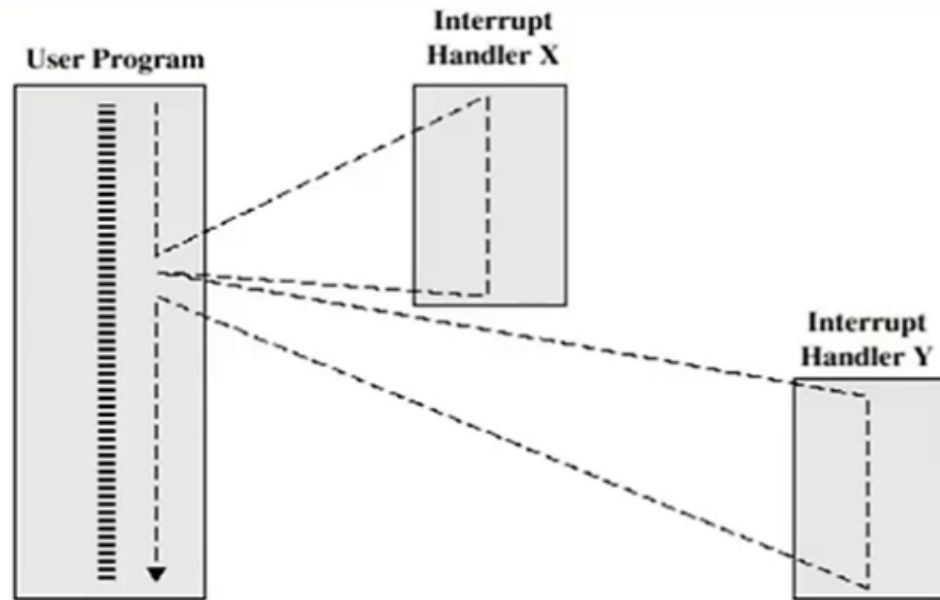
- **SCB (System Control Block):** Se encarga fundamentalmente del manejo excepciones del sistema.
  - Manejo de la tabla o vector de excepciones.
  - Habilitación e inhabilitación de excepciones.
  - Estado de las excepciones (inactiva, pendiente, activa)
  - Prioridades de las excepciones.
- **NVIC (Nested Vectored Interrupt Controller):** Se encarga de gestionar las interrupciones.
- **SysTick Timer:** Se encarga de gestionar el temporizador integrado en la arquitectura Cortex-M.

# Módulo NVIC

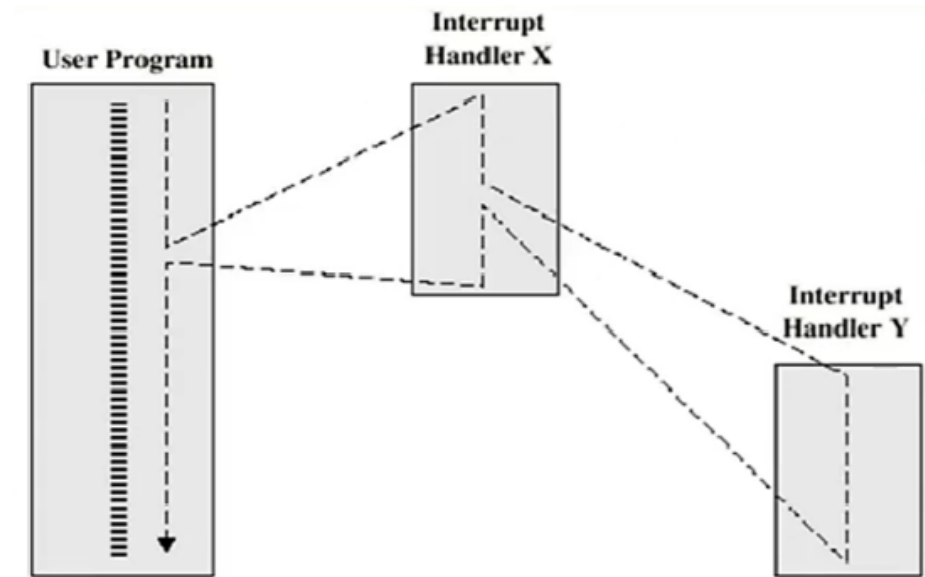
El NVIC (Nested Vectored Interrupt Control) es un Controlador de Interrupción Vectorial Anidado diseñado por ARM que se ubica dentro del core del microcontrolador.

## Concepto de Anidamiento

El NVIC permite atender interrupciones anidadas, es decir interrupciones adentro de otras interrupciones.



*Interrupcion Secuencial PIC*  
(Se puede atender solo una interrupción)



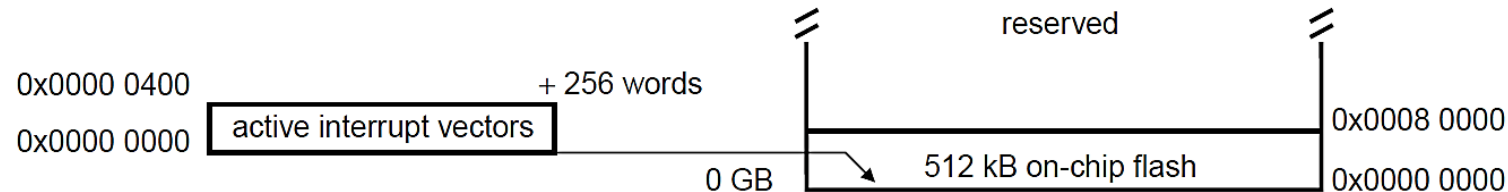
*Interrupcion Anidada ARM*  
(Se puede atender multiples interrupciones)

# Módulo NVIC

## Concepto de Vectorización

En ARM las interrupciones se manejan con una tabla de vectores de interrupción que contiene una matriz de punteros de función. Cada puntero corresponde con un vector de interrupción, en donde se coloca la dirección en donde estará la rutina de servicio a la interrupción (IRQs) o Handler. Cada una de las direcciones está asociada a un periférico determinado.

En ARM Cortex M la tabla de vectores de interrupción tiene 256 posiciones de memoria reubicables y seteadas de fábrica. Dado que cada palabra tiene 4 bytes, se obtiene un total de 64 vectores de interrupción.

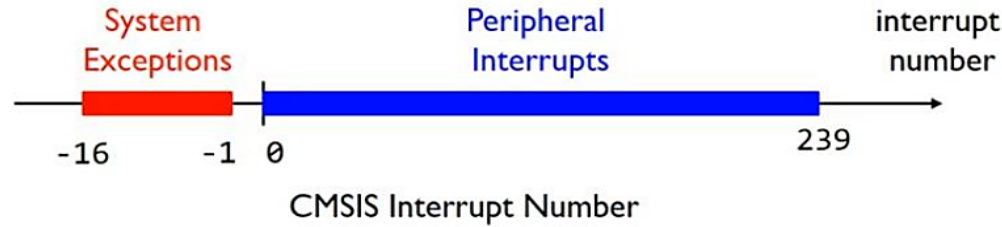


En ARM se tienen **excepciones** e **interrupciones**:

*Las **excepciones**, se tratan de eventos que ocurren internamente durante la ejecución del programa y pueden abarcar una gama más amplia de situaciones. Esto incluye condiciones anómalas como errores de ejecución, intentos de acceder a memoria no válida, divisiones por cero, y otros eventos que pueden requerir intervención especial.*

*Las **interrupciones**, se tratan específicamente de eventos externos generados por periféricos u otros dispositivos que requieren la atención inmediata del procesador. Estos eventos externos provocan la suspensión temporal de la ejecución normal del programa y la transferencia del control a una rutina de manejo de interrupción específica.*

# Módulo NVIC



En términos generales, la arquitectura Cortex-M3 admite hasta 256 vectores de interrupción y excepción. El NVIC del ARM Cortex M3 (ARM7RDMI) admite 35 interrupciones vectoriales (CMSIS considera las excepciones con número negativo y las interrupciones con número positivo).

**Nota:** Cada excepción tiene un nivel de prioridad, y sólo aquellas de mayor prioridad pueden interrumpir a otra que esté siendo atendida. Los números más bajos tienen precedencia, es decir, una prioridad más alta.

Las excepciones RESET, NMI y HARDFAULT tienen las prioridades más altas, fijas e inamovibles, -3, -2 y -1 respectivamente. El resto de las excepciones tiene su prioridad configurable por software mediante registros en el SCB o el NVIC, comenzando con el valor 0 (la prioridad más alta).

| Interrupt ID | Exception No. | Vector offset | Function       | Interrupt ID | Exception No. | Vector offset | Function              |
|--------------|---------------|---------------|----------------|--------------|---------------|---------------|-----------------------|
| 0            | 16            | 0x40          | Watchdog (WDT) | 18           | 34            | 0x88          | External interrupt    |
| 1            | 17            | 0x44          | Timer 0        | 19           | 35            | 0x8C          | External interrupt    |
| 2            | 18            | 0x48          | Timer 1        | 20           | 36            | 0x90          | External interrupt    |
| 3            | 19            | 0x4C          | Timer 2        | 21           | 37            | 0x94          | External interrupt    |
| 4            | 20            | 0x50          | Timer 3        | 22           | 38            | 0x98          | ADC                   |
| 5            | 21            | 0x54          | UART 0         | 23           | 39            | 0x9C          | BOD                   |
| 6            | 22            | 0x58          | UART 1         | 24           | 40            | 0xA0          | USB                   |
| 7            | 23            | 0x5C          | UART 2         | 25           | 41            | 0xA4          | CAN                   |
| 8            | 24            | 0x60          | UART 3         | 26           | 42            | 0xA8          | GPDMA                 |
| 9            | 25            | 0x64          | PWM 1          | 27           | 43            | 0xAC          | I2S                   |
| 10           | 26            | 0x68          | I2C0           | 28           | 44            | 0xB0          | Ethernet              |
| 11           | 27            | 0x6C          | I2C1           | 29           | 45            | 0xB4          | Repetitive int. timer |
| 12           | 28            | 0x70          | I2C2           | 30           | 46            | 0xB8          | PWM                   |
| 13           | 29            | 0x74          | SPI            | 31           | 47            | 0xBC          | Quadrature Encoder    |
| 14           | 30            | 0x78          | SSP0           | 32           | 48            | 0xC0          | PLL1 (USB PLL)        |
| 15           | 31            | 0x7C          | SSP1           | 33           | 49            | 0xC4          | USB activity int.     |
| 16           | 32            | 0x80          | PLL0           | 34           | 50            | 0xC8          | CAN activity int.     |
| 17           | 33            | 0x84          | RTC            |              |               |               |                       |

| Interrupt ID | Exception No. | Priority    | Vector offset | Function                                                              |
|--------------|---------------|-------------|---------------|-----------------------------------------------------------------------|
|              | 1             | -3, highest | 0x04          | Reset                                                                 |
| -14          | 2             | -2          | 0x08          | NMI                                                                   |
| -13          | 3             | -1          | 0x0C          | Hard fault :                                                          |
| -12          | 4             |             | 0x10          | Memory management fault : MPU violation , access to illegal locations |
| -11          | 5             |             | 0x14          | Bus fault : AHB interface receives an error response from slave       |
| -10          | 6             |             | 0x18          | Usage fault : Program error or trying to access coprocessor           |
|              | 7-10          |             |               | Reserved                                                              |
| -5           | 11            |             | 0x2C          | SVCALL : System Service call                                          |
|              | 12-13         |             |               | Reserved                                                              |
| -2           | 14            |             | 0x38          | PendSV : Pendable request for system device                           |
| -1           | 15            |             | 0x3C          | SysTick : System tick timer                                           |

# Módulo NVIC: Configuración Interrupciones por GPIO

Los siguientes registros configuran los pines del Puerto 0 y Puerto 2 para generar interrupciones.

## Registro IO0IntEnR (GPIO Interrupt Enable for port 0 Rising Edge)

Cada bit en estos registros habilita la interrupción del flanco ascendente para el correspondiente pin del puerto 0.

GPIO Interrupt Enable for port 0 Rising Edge (IO0IntEnR - 0x4002 8090) bit description

| Bit   | Symbol                | Value | Description                                | Reset value |
|-------|-----------------------|-------|--------------------------------------------|-------------|
| 0     | P0.0ER                |       | Enable rising edge interrupt for P0.0.     | 0           |
|       |                       | 0     | Rising edge interrupt is disabled on P0.0. |             |
|       |                       | 1     | Rising edge interrupt is enabled on P0.0.  |             |
| 1     | P0.1ER                |       | Enable rising edge interrupt for P0.1.     | 0           |
| 2     | P0.2ER                |       | Enable rising edge interrupt for P0.2.     | 0           |
| 3     | P0.3ER                |       | Enable rising edge interrupt for P0.3.     | 0           |
| 4     | P0.4ER <sup>[1]</sup> |       | Enable rising edge interrupt for P0.4.     | 0           |
| 5     | P0.5ER <sup>[1]</sup> |       | Enable rising edge interrupt for P0.5.     | 0           |
| 6     | P0.6ER                |       | Enable rising edge interrupt for P0.6.     | 0           |
| 7     | P0.7ER                |       | Enable rising edge interrupt for P0.7.     | 0           |
| 8     | P0.8ER                |       | Enable rising edge interrupt for P0.8.     | 0           |
| 9     | P0.9ER                |       | Enable rising edge interrupt for P0.9.     | 0           |
| 10    | P0.10ER               |       | Enable rising edge interrupt for P0.10.    | 0           |
| 11    | P0.11ER               |       | Enable rising edge interrupt for P0.11.    | 0           |
| 14:12 | -                     |       | Reserved                                   | NA          |
| 15    | P0.15ER               |       | Enable rising edge interrupt for P0.15.    | 0           |
| 16    | P0.16ER               |       | Enable rising edge interrupt for P0.16.    | 0           |

| Bit | Symbol                 | Value | Description                             | Reset value |
|-----|------------------------|-------|-----------------------------------------|-------------|
| 17  | P0.17ER                |       | Enable rising edge interrupt for P0.17. | 0           |
| 18  | P0.18ER                |       | Enable rising edge interrupt for P0.18. | 0           |
| 19  | P0.19ER <sup>[1]</sup> |       | Enable rising edge interrupt for P0.19. | 0           |
| 20  | P0.20ER <sup>[1]</sup> |       | Enable rising edge interrupt for P0.20. | 0           |
| 21  | P0.21ER <sup>[1]</sup> |       | Enable rising edge interrupt for P0.21. | 0           |
| 22  | P0.22ER                |       | Enable rising edge interrupt for P0.22. | 0           |
| 23  | P0.23ER <sup>[1]</sup> |       | Enable rising edge interrupt for P0.23. | 0           |
| 24  | P0.24ER <sup>[1]</sup> |       | Enable rising edge interrupt for P0.24. | 0           |
| 25  | P0.25ER                |       | Enable rising edge interrupt for P0.25. | 0           |
| 26  | P0.26ER                |       | Enable rising edge interrupt for P0.26. | 0           |
| 27  | P0.27ER <sup>[1]</sup> |       | Enable rising edge interrupt for P0.27. | 0           |
| 28  | P0.28ER <sup>[1]</sup> |       | Enable rising edge interrupt for P0.28. | 0           |
| 29  | P0.29ER                |       | Enable rising edge interrupt for P0.29. | 0           |
| 30  | P0.30ER                |       | Enable rising edge interrupt for P0.30. | 0           |
| 31  | -                      |       | Reserved.                               | NA          |

[1] Not available on 80-pin package.

# Módulo NVIC: Configuración Interrupciones por GPIO

## Registro IO2IntEnR (GPIO Interrupt Enable for port 2 Rising Edge)

Cada bit en estos registros habilita la interrupción del flanco ascendente para el correspondiente pin del puerto 2.

GPIO Interrupt Enable for port 2 Rising Edge (IO2IntEnR - 0x4002 80B0) bit description

| Bit   | Symbol                 | Value | Description                                | Reset value |
|-------|------------------------|-------|--------------------------------------------|-------------|
| 0     | P2.0ER                 |       | Enable rising edge interrupt for P2.0.     | 0           |
|       |                        | 0     | Rising edge interrupt is disabled on P2.0. |             |
|       |                        | 1     | Rising edge interrupt is enabled on P2.0.  |             |
| 1     | P2.1ER                 |       | Enable rising edge interrupt for P2.1.     | 0           |
| 2     | P2.2ER                 |       | Enable rising edge interrupt for P2.2.     | 0           |
| 3     | P2.3ER                 |       | Enable rising edge interrupt for P2.3.     | 0           |
| 4     | P2.4ER                 |       | Enable rising edge interrupt for P2.4.     | 0           |
| 5     | P2.5ER                 |       | Enable rising edge interrupt for P2.5.     | 0           |
| 6     | P2.6ER                 |       | Enable rising edge interrupt for P2.6.     | 0           |
| 7     | P2.7ER                 |       | Enable rising edge interrupt for P2.7.     | 0           |
| 8     | P2.8ER                 |       | Enable rising edge interrupt for P2.8.     | 0           |
| 9     | P2.9ER                 |       | Enable rising edge interrupt for P2.9.     | 0           |
| 10    | P2.10ER                |       | Enable rising edge interrupt for P2.10.    | 0           |
| 11    | P2.11ER <sup>[1]</sup> |       | Enable rising edge interrupt for P2.11.    | 0           |
| 12    | P2.12ER <sup>[1]</sup> |       | Enable rising edge interrupt for P2.12.    | 0           |
| 13    | P2.13ER <sup>[1]</sup> |       | Enable rising edge interrupt for P2.13.    | 0           |
| 31:14 | -                      |       | Reserved.                                  | NA          |

[1] Not available on 80-pin package.



# Módulo NVIC: Configuración Interrupciones por GPIO

## Registro IO0IntEnF (GPIO Interrupt Enable for port 0 Falling Edge)

Cada bit en estos registros habilita la interrupción del flanco descendente para el correspondiente pin del puerto 0.

GPIO Interrupt Enable for port 0 Falling Edge (IO0IntEnF - address 0x4002 8094) bit description

| Bit   | Symbol                | Value | Description                                 | Reset value |
|-------|-----------------------|-------|---------------------------------------------|-------------|
| 0     | P0.0EF                |       | Enable falling edge interrupt for P0.0      | 0           |
|       |                       | 0     | Falling edge interrupt is disabled on P0.0. |             |
|       |                       | 1     | Falling edge interrupt is enabled on P0.0.  |             |
| 1     | P0.1EF                |       | Enable falling edge interrupt for P0.1.     | 0           |
| 2     | P0.2EF                |       | Enable falling edge interrupt for P0.2.     | 0           |
| 3     | P0.3EF                |       | Enable falling edge interrupt for P0.3.     | 0           |
| 4     | P0.4EF <sup>[1]</sup> |       | Enable falling edge interrupt for P0.4.     | 0           |
| 5     | P0.5EF <sup>[1]</sup> |       | Enable falling edge interrupt for P0.5.     | 0           |
| 6     | P0.6EF                |       | Enable falling edge interrupt for P0.6.     | 0           |
| 7     | P0.7EF                |       | Enable falling edge interrupt for P0.7.     | 0           |
| 8     | P0.8EF                |       | Enable falling edge interrupt for P0.8.     | 0           |
| 9     | P0.9EF                |       | Enable falling edge interrupt for P0.9.     | 0           |
| 10    | P0.10EF               |       | Enable falling edge interrupt for P0.10.    | 0           |
| 11    | P0.11EF               |       | Enable falling edge interrupt for P0.11.    | 0           |
| 14:12 | -                     |       | Reserved.                                   | NA          |

| Bit | Symbol                 | Value | Description                              | Reset value |
|-----|------------------------|-------|------------------------------------------|-------------|
| 15  | P0.15EF                |       | Enable falling edge interrupt for P0.15. | 0           |
| 16  | P0.16EF                |       | Enable falling edge interrupt for P0.16. | 0           |
| 17  | P0.17EF                |       | Enable falling edge interrupt for P0.17. | 0           |
| 18  | P0.18EF                |       | Enable falling edge interrupt for P0.18. | 0           |
| 19  | P0.19EF <sup>[1]</sup> |       | Enable falling edge interrupt for P0.19. | 0           |
| 20  | P0.20EF <sup>[1]</sup> |       | Enable falling edge interrupt for P0.20. | 0           |
| 21  | P0.21EF <sup>[1]</sup> |       | Enable falling edge interrupt for P0.21. | 0           |
| 22  | P0.22EF                |       | Enable falling edge interrupt for P0.22. | 0           |
| 23  | P0.23EF <sup>[1]</sup> |       | Enable falling edge interrupt for P0.23. | 0           |
| 24  | P0.24EF <sup>[1]</sup> |       | Enable falling edge interrupt for P0.24. | 0           |
| 25  | P0.25EF                |       | Enable falling edge interrupt for P0.25. | 0           |
| 26  | P0.26EF                |       | Enable falling edge interrupt for P0.26. | 0           |
| 27  | P0.27EF <sup>[1]</sup> |       | Enable falling edge interrupt for P0.27. | 0           |
| 28  | P0.28EF <sup>[1]</sup> |       | Enable falling edge interrupt for P0.28. | 0           |
| 29  | P0.29EF                |       | Enable falling edge interrupt for P0.29. | 0           |
| 30  | P0.30EF                |       | Enable falling edge interrupt for P0.30. | 0           |
| 31  | -                      |       | Reserved.                                | NA          |

[1] Not available on 80-pin package.



# Módulo NVIC: Configuración Interrupciones por GPIO

## Registro IO2IntEnR (GPIO Interrupt Enable for port 2 Falling Edge)

Cada bit en estos registros habilita la interrupción del flanco descendente para el correspondiente pin del puerto 2.

GPIO Interrupt Enable for port 2 Falling Edge (IO2IntEnF - 0x4002 80B4) bit description

| Bit   | Symbol                 | Value | Description                                 | Reset value |
|-------|------------------------|-------|---------------------------------------------|-------------|
| 0     | P2.0EF                 |       | Enable falling edge interrupt for P2.0      | 0           |
|       |                        | 0     | Falling edge interrupt is disabled on P2.0. |             |
|       |                        | 1     | Falling edge interrupt is enabled on P2.0.  |             |
| 1     | P2.1EF                 |       | Enable falling edge interrupt for P2.1.     | 0           |
| 2     | P2.2EF                 |       | Enable falling edge interrupt for P2.2.     | 0           |
| 3     | P2.3EF                 |       | Enable falling edge interrupt for P2.3.     | 0           |
| 4     | P2.4EF                 |       | Enable falling edge interrupt for P2.4.     | 0           |
| 5     | P2.5EF                 |       | Enable falling edge interrupt for P2.5.     | 0           |
| 6     | P2.6EF                 |       | Enable falling edge interrupt for P2.6.     | 0           |
| 7     | P2.7EF                 |       | Enable falling edge interrupt for P2.7.     | 0           |
| 8     | P2.8EF                 |       | Enable falling edge interrupt for P2.8.     | 0           |
| 9     | P2.9EF                 |       | Enable falling edge interrupt for P2.9.     | 0           |
| 10    | P2.10EF                |       | Enable falling edge interrupt for P2.10.    | 0           |
| 11    | P2.11EF <sup>[1]</sup> |       | Enable falling edge interrupt for P2.11.    | 0           |
| 12    | P2.12EF <sup>[1]</sup> |       | Enable falling edge interrupt for P2.12.    | 0           |
| 13    | P2.13EF <sup>[1]</sup> |       | Enable falling edge interrupt for P2.13.    | 0           |
| 31:14 | -                      |       | Reserved.                                   | NA          |

[1] Not available on 80-pin package.

# Módulo NVIC: Configuración Interrupciones por GPIO

## Registro IO0IntStatR (GPIO Interrupt Status for port 0 Rising Edge Interrupt)

Cada bit en estos registros indica el estado de interrupción por flanco ascendente para el puerto 0.

GPIO Interrupt Status for port 0 Rising Edge Interrupt (IO0IntStatR - 0x4002 8084) bit description

| Bit   | Symbol                 | Value | Description                                                | Reset value |
|-------|------------------------|-------|------------------------------------------------------------|-------------|
| 0     | P0.0REI                |       | Status of Rising Edge Interrupt for P0.0                   | 0           |
|       |                        | 0     | A rising edge has not been detected on P0.0.               |             |
|       |                        | 1     | Interrupt has been generated due to a rising edge on P0.0. |             |
| 1     | P0.1REI                |       | Status of Rising Edge Interrupt for P0.1.                  | 0           |
| 2     | P0.2REI                |       | Status of Rising Edge Interrupt for P0.2.                  | 0           |
| 3     | P0.3REI                |       | Status of Rising Edge Interrupt for P0.3.                  | 0           |
| 4     | P0.4REI <sup>[1]</sup> |       | Status of Rising Edge Interrupt for P0.4.                  | 0           |
| 5     | P0.5REI <sup>[1]</sup> |       | Status of Rising Edge Interrupt for P0.5.                  | 0           |
| 6     | P0.6REI                |       | Status of Rising Edge Interrupt for P0.6.                  | 0           |
| 7     | P0.7REI                |       | Status of Rising Edge Interrupt for P0.7.                  | 0           |
| 8     | P0.8REI                |       | Status of Rising Edge Interrupt for P0.8.                  | 0           |
| 9     | P0.9REI                |       | Status of Rising Edge Interrupt for P0.9.                  | 0           |
| 10    | P0.10REI               |       | Status of Rising Edge Interrupt for P0.10.                 | 0           |
| 11    | P0.11REI               |       | Status of Rising Edge Interrupt for P0.11.                 | 0           |
| 14:12 | -                      |       | Reserved.                                                  | NA          |
| 15    | P0.15REI               |       | Status of Rising Edge Interrupt for P0.15.                 | 0           |

| Bit | Symbol                  | Value | Description                                | Reset value |
|-----|-------------------------|-------|--------------------------------------------|-------------|
| 16  | P0.16REI                |       | Status of Rising Edge Interrupt for P0.16. | 0           |
| 17  | P0.17REI                |       | Status of Rising Edge Interrupt for P0.17. | 0           |
| 18  | P0.18REI                |       | Status of Rising Edge Interrupt for P0.18. | 0           |
| 19  | P0.19REI <sup>[1]</sup> |       | Status of Rising Edge Interrupt for P0.19. | 0           |
| 20  | P0.20REI <sup>[1]</sup> |       | Status of Rising Edge Interrupt for P0.20. | 0           |
| 21  | P0.21REI <sup>[1]</sup> |       | Status of Rising Edge Interrupt for P0.21. | 0           |
| 22  | P0.22REI                |       | Status of Rising Edge Interrupt for P0.22. | 0           |
| 23  | P0.23REI <sup>[1]</sup> |       | Status of Rising Edge Interrupt for P0.23. | 0           |
| 24  | P0.24REI <sup>[1]</sup> |       | Status of Rising Edge Interrupt for P0.24. | 0           |
| 25  | P0.25REI                |       | Status of Rising Edge Interrupt for P0.25. | 0           |
| 26  | P0.26REI                |       | Status of Rising Edge Interrupt for P0.26. | 0           |
| 27  | P0.27REI <sup>[1]</sup> |       | Status of Rising Edge Interrupt for P0.27. | 0           |
| 28  | P0.28REI <sup>[1]</sup> |       | Status of Rising Edge Interrupt for P0.28. | 0           |
| 29  | P0.29REI                |       | Status of Rising Edge Interrupt for P0.29. | 0           |
| 30  | P0.30REI                |       | Status of Rising Edge Interrupt for P0.30. | 0           |
| 31  | -                       |       | Reserved.                                  | NA          |

[1] Not available on 80-pin package.

# Módulo NVIC: Configuración Interrupciones por GPIO

## Registro IO2IntStatR (GPIO Interrupt Status for port 2 Rising Edge Interrupt)

Cada bit en estos registros indica el estado de interrupción por flanco ascendente para el puerto 2.

GPIO Interrupt Status for port 2 Rising Edge Interrupt (IO2IntStatR - 0x4002 80A4) bit description

| Bit   | Symbol                  | Value | Description                                                | Reset value |
|-------|-------------------------|-------|------------------------------------------------------------|-------------|
| 0     | P2.0REI                 |       | Status of Rising Edge Interrupt for P2.0                   | 0           |
|       |                         | 0     | A rising edge has not been detected on P2.0.               |             |
|       |                         | 1     | Interrupt has been generated due to a rising edge on P2.0. |             |
| 1     | P2.1REI                 |       | Status of Rising Edge Interrupt for P2.1.                  | 0           |
| 2     | P2.2REI                 |       | Status of Rising Edge Interrupt for P2.2.                  | 0           |
| 3     | P2.3REI                 |       | Status of Rising Edge Interrupt for P2.3.                  | 0           |
| 4     | P2.4REI                 |       | Status of Rising Edge Interrupt for P2.4.                  | 0           |
| 5     | P2.5REI                 |       | Status of Rising Edge Interrupt for P2.5.                  | 0           |
| 6     | P2.6REI                 |       | Status of Rising Edge Interrupt for P2.6.                  | 0           |
| 7     | P2.7REI                 |       | Status of Rising Edge Interrupt for P2.7.                  | 0           |
| 8     | P2.8REI                 |       | Status of Rising Edge Interrupt for P2.8.                  | 0           |
| 9     | P2.9REI                 |       | Status of Rising Edge Interrupt for P2.9.                  | 0           |
| 10    | P2.10REI                |       | Status of Rising Edge Interrupt for P2.10.                 | 0           |
| 11    | P2.11REI <sup>[1]</sup> |       | Status of Rising Edge Interrupt for P2.11.                 | 0           |
| 12    | P2.12REI <sup>[1]</sup> |       | Status of Rising Edge Interrupt for P2.12.                 | 0           |
| 13    | P2.13REI <sup>[1]</sup> |       | Status of Rising Edge Interrupt for P2.13.                 | 0           |
| 31:14 | -                       |       | Reserved.                                                  | NA          |

[1] Not available on 80-pin package.

# Módulo NVIC: Configuración Interrupciones por GPIO

## Registro IO0IntStatF (GPIO Interrupt Status for port 0 Falling Edge Interrupt)

Cada bit en estos registros indica el estado de interrupción por flanco descendente para el puerto 0.

GPIO Interrupt Status for port 0 Falling Edge Interrupt (IO0IntStatF - 0x4002 8088) bit description

| Bit   | Symbol                 | Value | Description                                                 | Reset value |
|-------|------------------------|-------|-------------------------------------------------------------|-------------|
| 0     | P0.0FEI                |       | Status of Falling Edge Interrupt for P0.0                   | 0           |
|       |                        | 0     | A falling edge has not been detected on P0.0.               |             |
|       |                        | 1     | Interrupt has been generated due to a falling edge on P0.0. |             |
| 1     | P0.1FEI                |       | Status of Falling Edge Interrupt for P0.1.                  | 0           |
| 2     | P0.2FEI                |       | Status of Falling Edge Interrupt for P0.2.                  | 0           |
| 3     | P0.3FEI                |       | Status of Falling Edge Interrupt for P0.3.                  | 0           |
| 4     | P0.4FEI <sup>[1]</sup> |       | Status of Falling Edge Interrupt for P0.4.                  | 0           |
| 5     | P0.5FEI <sup>[1]</sup> |       | Status of Falling Edge Interrupt for P0.5.                  | 0           |
| 6     | P0.6FEI                |       | Status of Falling Edge Interrupt for P0.6.                  | 0           |
| 7     | P0.7FEI                |       | Status of Falling Edge Interrupt for P0.7.                  | 0           |
| 8     | P0.8FEI                |       | Status of Falling Edge Interrupt for P0.8.                  | 0           |
| 9     | P0.9FEI                |       | Status of Falling Edge Interrupt for P0.9.                  | 0           |
| 10    | P0.10FEI               |       | Status of Falling Edge Interrupt for P0.10.                 | 0           |
| 11    | P0.11FEI               |       | Status of Falling Edge Interrupt for P0.11.                 | 0           |
| 14:12 | -                      |       | Reserved.                                                   | NA          |
| 15    | P0.15FEI               |       | Status of Falling Edge Interrupt for P0.15.                 | 0           |

| Bit | Symbol                  | Value | Description                                 | Reset value |
|-----|-------------------------|-------|---------------------------------------------|-------------|
| 16  | P0.16FEI                |       | Status of Falling Edge Interrupt for P0.16. | 0           |
| 17  | P0.17FEI                |       | Status of Falling Edge Interrupt for P0.17. | 0           |
| 18  | P0.18FEI                |       | Status of Falling Edge Interrupt for P0.18. | 0           |
| 19  | P0.19FEI <sup>[1]</sup> |       | Status of Falling Edge Interrupt for P0.19. | 0           |
| 20  | P0.20FEI <sup>[1]</sup> |       | Status of Falling Edge Interrupt for P0.20. | 0           |
| 21  | P0.21FEI <sup>[1]</sup> |       | Status of Falling Edge Interrupt for P0.21. | 0           |
| 22  | P0.22FEI                |       | Status of Falling Edge Interrupt for P0.22. | 0           |
| 23  | P0.23FEI <sup>[1]</sup> |       | Status of Falling Edge Interrupt for P0.23. | 0           |
| 24  | P0.24FEI <sup>[1]</sup> |       | Status of Falling Edge Interrupt for P0.24. | 0           |
| 25  | P0.25FEI                |       | Status of Falling Edge Interrupt for P0.25. | 0           |
| 26  | P0.26FEI                |       | Status of Falling Edge Interrupt for P0.26. | 0           |
| 27  | P0.27FEI <sup>[1]</sup> |       | Status of Falling Edge Interrupt for P0.27. | 0           |
| 28  | P0.28FEI <sup>[1]</sup> |       | Status of Falling Edge Interrupt for P0.28. | 0           |
| 29  | P0.29FEI                |       | Status of Falling Edge Interrupt for P0.29. | 0           |
| 30  | P0.30FEI                |       | Status of Falling Edge Interrupt for P0.30. | 0           |
| 31  | -                       |       | Reserved.                                   | NA          |

[1] Not available on 80-pin package.

# Módulo NVIC: Configuración Interrupciones por GPIO

## Registro IO2IntStatF (GPIO Interrupt Status for port 2 Falling Edge Interrupt)

Cada bit en estos registros indica el estado de interrupción por flanco descendente para el puerto 2.

GPIO Interrupt Status for port 2 Falling Edge Interrupt (IO2IntStatF - 0x4002 80A8) bit description

| Bit   | Symbol                  | Value | Description                                                 | Reset value |
|-------|-------------------------|-------|-------------------------------------------------------------|-------------|
| 0     | P2.0FEI                 |       | Status of Falling Edge Interrupt for P2.0                   | 0           |
|       |                         | 0     | A falling edge has not been detected on P2.0.               |             |
|       |                         | 1     | Interrupt has been generated due to a falling edge on P2.0. |             |
| 1     | P2.1FEI                 |       | Status of Falling Edge Interrupt for P2.1.                  | 0           |
| 2     | P2.2FEI                 |       | Status of Falling Edge Interrupt for P2.2.                  | 0           |
| 3     | P2.3FEI                 |       | Status of Falling Edge Interrupt for P2.3.                  | 0           |
| 4     | P2.4FEI                 |       | Status of Falling Edge Interrupt for P2.4.                  | 0           |
| 5     | P2.5FEI                 |       | Status of Falling Edge Interrupt for P2.5.                  | 0           |
| 6     | P2.6FEI                 |       | Status of Falling Edge Interrupt for P2.6.                  | 0           |
| 7     | P2.7FEI                 |       | Status of Falling Edge Interrupt for P2.7.                  | 0           |
| 8     | P2.8FEI                 |       | Status of Falling Edge Interrupt for P2.8.                  | 0           |
| 9     | P2.9FEI                 |       | Status of Falling Edge Interrupt for P2.9.                  | 0           |
| 10    | P2.10FEI                |       | Status of Falling Edge Interrupt for P2.10.                 | 0           |
| 11    | P2.11FEI <sup>[1]</sup> |       | Status of Falling Edge Interrupt for P2.11.                 | 0           |
| 12    | P2.12FEI <sup>[1]</sup> |       | Status of Falling Edge Interrupt for P2.12.                 | 0           |
| 13    | P2.13FEI <sup>[1]</sup> |       | Status of Falling Edge Interrupt for P2.13.                 | 0           |
| 31:14 | -                       |       | Reserved.                                                   | NA          |

[1] Not available on 80-pin package.



# Módulo NVIC: Configuración Interrupciones por GPIO

## Registro IO0IntClr (GPIO Interrupt Clear register for port 0)

Escribir un 1 en un bit en este registro borra cualquier interrupción para el correspondiente pin del puerto 0.

GPIO Interrupt Clear register for port 0 (IO0IntClr - 0x4002 808C)) bit description

| Bit   | Symbol                | Value | Description                                                      | Reset value |
|-------|-----------------------|-------|------------------------------------------------------------------|-------------|
| 0     | P0.0CI                |       | Clear GPIO port Interrupts for P0.0                              | 0           |
|       |                       | 0     | Corresponding bits in IOxIntStatR and IOxIntStatF are unchanged. |             |
|       |                       | 1     | Corresponding bits in IOxIntStatR and IOxStatF are cleared.      |             |
| 1     | P0.1CI                |       | Clear GPIO port Interrupts for P0.1.                             | 0           |
| 2     | P0.2CI                |       | Clear GPIO port Interrupts for P0.2.                             | 0           |
| 3     | P0.3CI                |       | Clear GPIO port Interrupts for P0.3.                             | 0           |
| 4     | P0.4CI <sup>[1]</sup> |       | Clear GPIO port Interrupts for P0.4.                             | 0           |
| 5     | P0.5CI <sup>[1]</sup> |       | Clear GPIO port Interrupts for P0.5.                             | 0           |
| 6     | P0.6CI                |       | Clear GPIO port Interrupts for P0.6.                             | 0           |
| 7     | P0.7CI                |       | Clear GPIO port Interrupts for P0.7.                             | 0           |
| 8     | P0.8CI                |       | Clear GPIO port Interrupts for P0.8.                             | 0           |
| 9     | P0.9CI                |       | Clear GPIO port Interrupts for P0.9.                             | 0           |
| 10    | P0.10CI               |       | Clear GPIO port Interrupts for P0.10.                            | 0           |
| 11    | P0.11CI               |       | Clear GPIO port Interrupts for P0.11.                            | 0           |
| 14:12 | -                     |       | Reserved.                                                        | NA          |
| 15    | P0.15CI               |       | Clear GPIO port Interrupts for P0.15.                            | 0           |

| Bit | Symbol                 | Value | Description                           | Reset value |
|-----|------------------------|-------|---------------------------------------|-------------|
| 16  | P0.16CI                |       | Clear GPIO port Interrupts for P0.16. | 0           |
| 17  | P0.17CI                |       | Clear GPIO port Interrupts for P0.17. | 0           |
| 18  | P0.18CI                |       | Clear GPIO port Interrupts for P0.18. | 0           |
| 19  | P0.19CI <sup>[1]</sup> |       | Clear GPIO port Interrupts for P0.19. | 0           |
| 20  | P0.20CI <sup>[1]</sup> |       | Clear GPIO port Interrupts for P0.20. | 0           |
| 21  | P0.21CI <sup>[1]</sup> |       | Clear GPIO port Interrupts for P0.21. | 0           |
| 22  | P0.22CI                |       | Clear GPIO port Interrupts for P0.22. | 0           |
| 23  | P0.23CI <sup>[1]</sup> |       | Clear GPIO port Interrupts for P0.23. | 0           |
| 24  | P0.24CI <sup>[1]</sup> |       | Clear GPIO port Interrupts for P0.24. | 0           |
| 25  | P0.25CI                |       | Clear GPIO port Interrupts for P0.25. | 0           |
| 26  | P0.26CI                |       | Clear GPIO port Interrupts for P0.26. | 0           |
| 27  | P0.27CI <sup>[1]</sup> |       | Clear GPIO port Interrupts for P0.27. | 0           |
| 28  | P0.28CI <sup>[1]</sup> |       | Clear GPIO port Interrupts for P0.28. | 0           |
| 29  | P0.29CI                |       | Clear GPIO port Interrupts for P0.29. | 0           |
| 30  | P0.30CI                |       | Clear GPIO port Interrupts for P0.30. | 0           |
| 31  | -                      |       | Reserved.                             | NA          |

[1] Not available on 80-pin package.

# Módulo NVIC: Configuración Interrupciones por GPIO

## Registro IO2IntClr (GPIO Interrupt Clear register for port 2)

Escribir un 1 en un bit en este registro borra cualquier interrupción para el correspondiente pin del puerto 2.

| GPIO Interrupt Clear register for port 2 (IO2IntClr - 0x4002 80AC) bit description |                        |       |                                                                  |             |
|------------------------------------------------------------------------------------|------------------------|-------|------------------------------------------------------------------|-------------|
| Bit                                                                                | Symbol                 | Value | Description                                                      | Reset value |
| 0                                                                                  | P2.0CI                 |       | Clear GPIO port Interrupts for P2.0                              | 0           |
|                                                                                    |                        | 0     | Corresponding bits in IOxIntStatR and IOxIntStatF are unchanged. |             |
|                                                                                    |                        | 1     | Corresponding bits in IOxIntStatR and IOxStatF are cleared.      |             |
| 1                                                                                  | P2.1CI                 |       | Clear GPIO port Interrupts for P2.1.                             | 0           |
| 2                                                                                  | P2.2CI                 |       | Clear GPIO port Interrupts for P2.2.                             | 0           |
| 3                                                                                  | P2.3CI                 |       | Clear GPIO port Interrupts for P2.3.                             | 0           |
| 4                                                                                  | P2.4CI                 |       | Clear GPIO port Interrupts for P2.4.                             | 0           |
| 5                                                                                  | P2.5CI                 |       | Clear GPIO port Interrupts for P2.5.                             | 0           |
| 6                                                                                  | P2.6CI                 |       | Clear GPIO port Interrupts for P2.6.                             | 0           |
| 7                                                                                  | P2.7CI                 |       | Clear GPIO port Interrupts for P2.7.                             | 0           |
| 8                                                                                  | P2.8CI                 |       | Clear GPIO port Interrupts for P2.8.                             | 0           |
| 9                                                                                  | P2.9CI                 |       | Clear GPIO port Interrupts for P2.9.                             | 0           |
| 10                                                                                 | P2.10CI                |       | Clear GPIO port Interrupts for P2.10.                            | 0           |
| 11                                                                                 | P2.11CI <sup>[1]</sup> |       | Clear GPIO port Interrupts for P2.11.                            | 0           |
| 12                                                                                 | P2.12CI <sup>[1]</sup> |       | Clear GPIO port Interrupts for P2.12.                            | 0           |
| 13                                                                                 | P2.13CI <sup>[1]</sup> |       | Clear GPIO port Interrupts for P2.13.                            | 0           |
| 31:14                                                                              | -                      |       | Reserved.                                                        | NA          |

[1] Not available on 80-pin package.



# Módulo NVIC: Configuración Interrupciones por GPIO

## Registro IOIntStatus (GPIO overall Interrupt Status register)

Este registro indica la presencia de interrupción pendiente en todos los puertos que admiten interrupciones GPIO. Solo se requiere un bit de estado por puerto.

GPIO overall Interrupt Status register (IOIntStatus - address 0x4002 8080) bit description

| Bit  | Symbol | Value | Description                                                  | Reset value |
|------|--------|-------|--------------------------------------------------------------|-------------|
| 0    | P0Int  |       | Port 0 GPIO interrupt pending.                               | 0           |
|      |        | 0     | There are no pending interrupts on Port 0.                   |             |
|      |        | 1     | There is at least one pending interrupt on Port 0.           |             |
| 1    | -      | -     | Reserved. The value read from a reserved bit is not defined. | NA          |
| 2    | P2Int  |       | Port 2 GPIO interrupt pending.                               | 0           |
|      |        | 0     | There are no pending interrupts on Port 2.                   |             |
|      |        | 1     | There is at least one pending interrupt on Port 2.           |             |
| 31:2 | -      | -     | Reserved. The value read from a reserved bit is not defined. | NA          |

# Módulo NVIC: Configuración de Interrupciones Externas

## Registro EXTINT (External Interrupt Flag register)

External Interrupt Flag register (EXTINT - address 0x400F C140) bit description

| Bit | Symbol | Description                                                                                                                                                                                                                                                                                                                                                                                                  | Reset value |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 0   | EINT0  | <p>In level-sensitive mode, this bit is set if the EINT0 function is selected for its pin, and the pin is in its active state. In edge-sensitive mode, this bit is set if the EINT0 function is selected for its pin, and the selected edge occurs on the pin.</p> <p>This bit is cleared by writing a one to it, except in level sensitive mode when the pin is in its active state.<a href="#">[1]</a></p> | 0           |
| 1   | EINT1  | <p>In level-sensitive mode, this bit is set if the EINT1 function is selected for its pin, and the pin is in its active state. In edge-sensitive mode, this bit is set if the EINT1 function is selected for its pin, and the selected edge occurs on the pin.</p> <p>This bit is cleared by writing a one to it, except in level sensitive mode when the pin is in its active state.<a href="#">[1]</a></p> | 0           |

| Bit  | Symbol | Description                                                                                                                                                                                                                                                                                                                                                                                                  | Reset value |
|------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 2    | EINT2  | <p>In level-sensitive mode, this bit is set if the EINT2 function is selected for its pin, and the pin is in its active state. In edge-sensitive mode, this bit is set if the EINT2 function is selected for its pin, and the selected edge occurs on the pin.</p> <p>This bit is cleared by writing a one to it, except in level sensitive mode when the pin is in its active state.<a href="#">[1]</a></p> | 0           |
| 3    | EINT3  | <p>In level-sensitive mode, this bit is set if the EINT3 function is selected for its pin, and the pin is in its active state. In edge-sensitive mode, this bit is set if the EINT3 function is selected for its pin, and the selected edge occurs on the pin.</p> <p>This bit is cleared by writing a one to it, except in level sensitive mode when the pin is in its active state.<a href="#">[1]</a></p> | 0           |
| 31:4 | -      | Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.                                                                                                                                                                                                                                                                                           | NA          |

[1] Example: e.g. if the EINTx is selected to be low level sensitive and low level is present on corresponding pin, this bit can not be cleared; this bit can be cleared only when signal on the pin becomes high.

# Módulo NVIC: Configuración de Interrupciones Externas

## Registro EXTMODE (External Interrupt Mode register)

Los bits en este registro seleccionan si cada pin EINT es sensible a una interrupción por nivel o por flanco.

External Interrupt Mode register (EXTMODE - address 0x400F C148) bit description

| Bit  | Symbol   | Value | Description                                                                                                        | Reset value |
|------|----------|-------|--------------------------------------------------------------------------------------------------------------------|-------------|
| 0    | EXTMODE0 | 0     | Level-sensitivity is selected for $\overline{\text{EINT0}}$ .                                                      | 0           |
|      |          | 1     | $\overline{\text{EINT0}}$ is edge sensitive.                                                                       |             |
| 1    | EXTMODE1 | 0     | Level-sensitivity is selected for $\overline{\text{EINT1}}$ .                                                      | 0           |
|      |          | 1     | $\overline{\text{EINT1}}$ is edge sensitive.                                                                       |             |
| 2    | EXTMODE2 | 0     | Level-sensitivity is selected for $\overline{\text{EINT2}}$ .                                                      | 0           |
|      |          | 1     | $\overline{\text{EINT2}}$ is edge sensitive.                                                                       |             |
| 3    | EXTMODE3 | 0     | Level-sensitivity is selected for $\overline{\text{EINT3}}$ .                                                      | 0           |
|      |          | 1     | $\overline{\text{EINT3}}$ is edge sensitive.                                                                       |             |
| 31:4 | -        | -     | Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined. | NA          |

# Módulo NVIC: Configuración de Interrupciones Externas

## Registro EXTPOLAR (External Interrupt Polarity register)

En el modo sensible por nivel, los bits en este registro seleccionan si el pin correspondiente es activo por alto o bajo. En el modo sensible por flanco, seleccionan si el pin es activo por flanco ascendente o por flanco descendente.

External Interrupt Polarity register (EXTPOLAR - address 0x400F C14C) bit description

| Bit  | Symbol    | Value | Description                                                                                                        | Reset value |
|------|-----------|-------|--------------------------------------------------------------------------------------------------------------------|-------------|
| 0    | EXTPOLAR0 | 0     | $\overline{\text{EINT0}}$ is low-active or falling-edge sensitive (depending on EXTMODE0).                         | 0           |
|      |           | 1     | $\overline{\text{EINT0}}$ is high-active or rising-edge sensitive (depending on EXTMODE0).                         |             |
| 1    | EXTPOLAR1 | 0     | $\overline{\text{EINT1}}$ is low-active or falling-edge sensitive (depending on EXTMODE1).                         | 0           |
|      |           | 1     | $\overline{\text{EINT1}}$ is high-active or rising-edge sensitive (depending on EXTMODE1).                         |             |
| 2    | EXTPOLAR2 | 0     | $\overline{\text{EINT2}}$ is low-active or falling-edge sensitive (depending on EXTMODE2).                         | 0           |
|      |           | 1     | $\overline{\text{EINT2}}$ is high-active or rising-edge sensitive (depending on EXTMODE2).                         |             |
| 3    | EXTPOLAR3 | 0     | $\overline{\text{EINT3}}$ is low-active or falling-edge sensitive (depending on EXTMODE3).                         | 0           |
|      |           | 1     | $\overline{\text{EINT3}}$ is high-active or rising-edge sensitive (depending on EXTMODE3).                         |             |
| 31:4 | -         | -     | Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined. | NA          |

# Módulo NVIC: Funciones de Control de Interrupciones por CMSIS

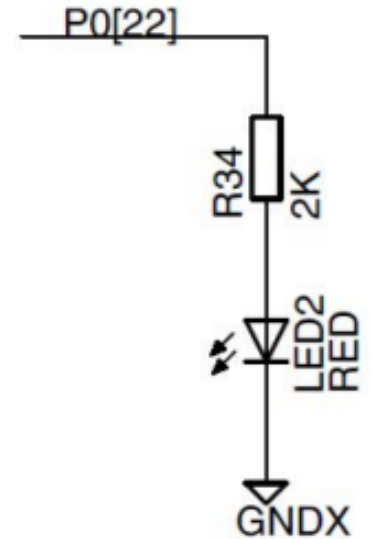
Las funciones para configurar interrupciones son:

## CMSIS functions for NVIC control

| CMSIS interrupt control function                                       | Description                                   |
|------------------------------------------------------------------------|-----------------------------------------------|
| <code>void NVIC_SetPriorityGrouping(uint32_t priority_grouping)</code> | Set the priority grouping                     |
| <code>void NVIC_EnableIRQ(IRQn_t IRQn)</code>                          | Enable IRQn                                   |
| <code>void NVIC_DisableIRQ(IRQn_t IRQn)</code>                         | Disable IRQn                                  |
| <code>uint32_t NVIC_GetPendingIRQ (IRQn_t IRQn)</code>                 | Return true (IRQ-Number) if IRQn is pending   |
| <code>void NVIC_SetPendingIRQ (IRQn_t IRQn)</code>                     | Set IRQn pending                              |
| <code>void NVIC_ClearPendingIRQ (IRQn_t IRQn)</code>                   | Clear IRQn pending status                     |
| <code>uint32_t NVIC_GetActive (IRQn_t IRQn)</code>                     | Return the IRQ number of the active interrupt |
| <code>void NVIC_SetPriority (IRQn_t IRQn, uint32_t priority)</code>    | Set priority for IRQn                         |
| <code>uint32_t NVIC_GetPriority (IRQn_t IRQn)</code>                   | Read priority of IRQn                         |
| <code>void NVIC_SystemReset (void)</code>                              | Reset the system                              |

# Ejemplo 1: Interrupción por GPIO

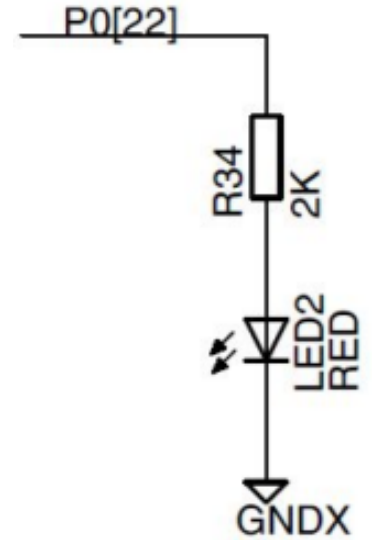
```
1  /*
2   * @brief  Enciende y apaga un led conectado al pin P0.22 con un periodo de
3   *          intermitencia dado por el pulsador ubicado en el pin P0.15.
4   */
5
6  /* Directivas de Inclusión */
7  #include"LPC17xx.h"
8
9  /* Definición de Variables Globales */
10 uint8_t inte = 0;
11
12 /* Prototipos de Funciones */
13 void cfgGPIO(void);
14 void cfgIntGPIO(void);
15 void delay(uint32_t time);
16
17 /* Función Principal */
18 int main(void){
19
20     uint32_t time;
21
22     cfgGPIO();
23     cfgIntGPIO();
24
25     while(1)
26     {
27         if(inte%2){
28             time = 1000000;
29         }
30         else {
31             time = 4000000;
32         }
33
34         LPC_GPIO0->FIOSET = (1<<22);
35         retardo(time);
36         LPC_GPIO0->FIOCLR = (1<<22);
37         retardo(time);
38     }
39     return 0;
40 }
```





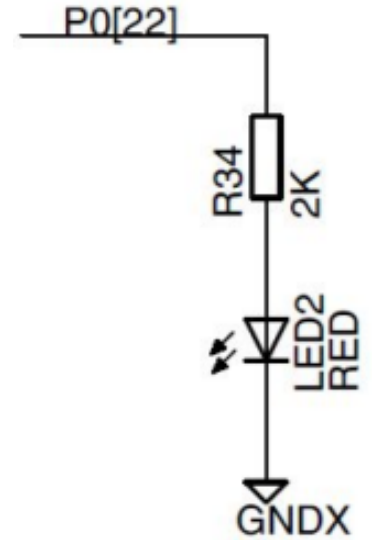
# Ejemplo 1: Interrupción por GPIO

```
41
42  /* Definiciones de Funciones */
43  void cfgGPIO(void)
44  {
45      LPC_PINCON->PINSEL0  &= ~(0b11<<30); // P0.15 GPIO Function
46      LPC_PINCON->PINSEL1  &= ~(0b11<<12); // P0.22 GPIO Function
47      LPC_GPIO0->FIODIR    &= ~(1<<15);    // P0.15 Digital Input
48      LPC_GPIO0->FIODIR    |= (1<<22);      // P0.22 Digital Output
49      return;
50  }
51
52  void cfgIntGPIO(void)
53  {
54      LPC_GPIOINT->IO0IntEnR |= (1<<15); //INTGPIO P0.15 por Flanco Ascendente
55      LPC_GPIOINT->IO0IntClr |= (1<<15); //Limpia Flag INTGPIO P0.15
56      NVIC_SetPriority(EINT3_IRQn,1);    //Prioridad 1 para Interrupción Externa EINT3
57      NVIC_EnableIRQ(EINT3_IRQn);        //Habilitación de Interrupción Externa EINT3
58      return;
59  }
60
61  void EINT3_IRQHandler(void)
62  {
63      //Consulta Puerto de Interrupción
64      if(LPC_GPIOINT->IntStatus & (1<<0)){ //Se produjo una interrupción en P0?
65          //Consulta Pin de Interrupción
66          if(LPC_GPIOINT->IO0IntStatR & (1<<15)){ //Se produjo una interrupción en P0.15?
67              inte++;
68              LPC_GPIOINT->IO0IntClr |= (1<<15); //Limpia Flag INTGPIO P0.15
69          }
70      }
71      return;
72  }
73
74  void delay(uint32_t time)
75  {
76      uint32_t counter;
77      for(counter = 0 ; counter < time; counter++){;}
78  }
```



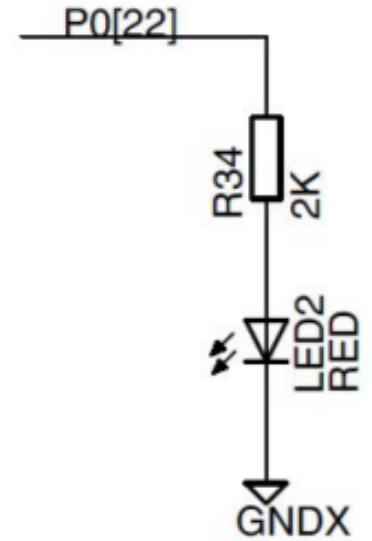
# Ejemplo 2: Interrupción Externa

```
1  /*
2   * @brief   Enciende y apaga un led conectado al pin P0.22 con un periodo de
3   *          intermitencia dado por el pulsador ubicado en el pin P2.10.
4   */
5
6  /* Directivas de Inclusión */
7  #include "LPC17xx.h"
8
9  /* Definición de Variables Globales */
10 uint8_t inte = 0;
11
12 /* Prototipos de Funciones */
13 void cfgGPIO(void);
14 void cfgIntExt(void);
15 void delay(uint32_t time);
16
17 /* Función Principal */
18 int main(void){
19
20     uint32_t time;
21
22     cfgGPIO();
23     cfgIntExt();
24
25     while(1)
26     {
27         if(inte%2){
28             time = 1000000;
29         }
30         else {
31             time = 4000000;
32         }
33
34         LPC_GPIO0->FIOSET = (1<<22);
35         retardo(time);
36         LPC_GPIO0->FIOCLR = (1<<22);
37         retardo(time);
38     }
39     return 0;
40 }
```



# Ejemplo 2: Interrupción Externa

```
41
42  /* Definiciones de Funciones */
43  void cfgGPIO(void)
44  {
45      LPC_PINCON->PINSEL4 |= (1<<20);    //P0.10 EINT0
46      LPC_PINCON->PINSEL1 &= ~(0b11<<12); //P0.22 GPIO Function
47      LPC_GPIO0->FIODIR  |= (1<<22);    //P0.22 Digital Output
48      return;
49  }
50
51  void cfgIntExt(void)
52  {
53      LPC_SC->EXTMODE  |= (1<<0);    //EINT0 por Flanco
54      LPC_SC->EXTPOLAR |= (1<<0);    //EINT0 por Flanco Ascendente
55      LPC_SC->EXTINT   |= (1<<0);    //Limpia Flag EINT0
56      NVIC_SetPriority(EINT0_IRQn,3); //Prioridad 3 para Interrupción Externa EINT0
57      NVIC_EnableIRQ(EINT0_IRQn);   //Habilitación de Interrupción Externa EINT0
58      return;
59  }
60
61  void EINT0_IRQHandler(void)
62  {
63      inte++;
64      LPC_SC->EXTINT |= (1<<0);    //Limpia Flag EINT0
65      return;
66  }
67
68  void delay(uint32_t time)
69  {
70      uint32_t counter;
71      for(counter = 0 ; counter < time; counter++){};
72  }
```



## **Ejercicio 1: Interrupción por GPIO**

Realizar un programa que configure el puerto P2.0 y P2.1 para que provoquen una interrupción por flanco de subida para el primer pin y por flanco de bajada para el segundo. Cuando la interrupción sea por P2.0 se enviará por el pin P0.0 la secuencia de bits 010011010. Si la interrupción es por P2.1 se enviará por el pin P0.1 la secuencia 011100110. Las secuencias se envían únicamente cuando se produce una interrupción, en caso contrario la salida de los pines tienen valores 1 lógicos. ¿que prioridad tienen configuradas por defecto estas interrupciones?

## **Ejercicio 2: Interrupción por GPIO**

Realizar un programa que configure el puerto P0.0 y P2.0 para que provoquen una interrupción por flanco de subida. Si la interrupción es por P0.0 guardar el valor binario 100111 en la variable "auxiliar", si es por P2.0 guardar el valor binario 111001011010110.

### **Ejercicio 3: Interrupción Externa**

Configurar la interrupción externa EINT1 para que interrumpa por flanco de bajada y la interrupción EINT2 para que interrumpa por flanco de subida. En la interrupción por flanco de bajada configurar el systick para desbordar cada 25 mseg, mientras que en la interrupción por flanco de subida configurarlo para que desborde cada 60 mseg. Considerar que EINT1 tiene mayor prioridad que EINT2.



## **Ejercicio 4: Interrupción Externa**

Escribir un programa en C que permita sacar por los pines P1.16 al P1.23 el equivalente en ascii de "a" si la interrupción se ha realizado por interrupción de EINT1 y no hay interrupción pendiente de EINT0. Para el caso en el cual se produzca una interrupción por EINT1 y exista una interrupción pendiente de EINT0 sacar por el puerto el equivalente en ascii de "A". La interrupción de EINT1 es por el flanco de subida producido por un pulsador identificado como "tecla a/A", mientras que la interrupción EINT0 es por el nivel alto de un pulsador etiquetado como "Activación de Mayúsculas".

Nota: Valerse de la configuración de los niveles de prioridad para que la pulsación conjunta de "Activación de Mayúsculas" con "tecla a/A" de como resultado a la salida el equivalente en ascii "A".